Quasi-inverses are not a mere proposition: an explicit counterexample

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'thm:qinv-notprop' (source lines 120-122); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the auto-homotopy lemma, whose hypotheses the proof checks one by one and whose item labels it refers to; retaining it keeps those labels defined inside the package.

Editorial note. The retained counter values make the counted result print with its original book number, and the chapter and section counters are set so that the retained context (where present) prints with its own original number as well.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: quasi-inverses are not a mere proposition (author label thm:qinv-notprop).

Allowed standard prerequisites (editorial).

- Automatic-homotopy lemma. If a = a is a set, every x is merely equal to a and q commutes with all self-paths, then there is f : prd x:X (x = x) with f(a) = q.. It is applied to the type X and the point a at source line 129.
- No higher axiom of choice. The type Sigma A:type, brck(bool = A) is used earlier in the book to refute a higher form of the axiom of choice.. The same type is reused here at source line 125.
- Equivalences of the booleans. Every equivalence bool ~ bool is equal to the identity or to the swapping equivalence.. It is invoked at source line 132 for the four-way case analysis.

Source reference. Original theorem 3.8.5: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1833–1833. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original section 3.5: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1735–1735. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Source reference. Original exercise 2.13: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1356–1356. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Lemma 4.1.2. *Suppose we have a type A with $a : A$ and $q : a = a$ such that*

(i) The type $a = a$ is a set.
(ii) For all $x : A$ we have $\|a = x\|$.
(iii) For all $p : a = a$ we have $p \cdot q = q \cdot p$.

Then there exists $f : \prod_{(x:A)}(x = x)$ with $f(a) = q$.

Theorem 4.1.3. *There exist types A and B and a function $f : A \to B$ such that $\mathsf{qinv}(f)$ is not a mere proposition.*

Proof. It suffices to exhibit a type X such that $\prod_{(x:X)}(x = x)$ is not a mere proposition. Define $X :\equiv \sum_{(A:\mathcal{U})}\|\mathbf{2} = A\|$, as in the proof of original theorem 3.8.5. It will suffice to exhibit an $f : \prod_{(x:X)}(x = x)$ which is unequal to $\lambda x.\, \mathsf{refl}_x$.

Let $a :\equiv (\mathbf{2}, |\mathsf{refl}_{\mathbf{2}}|) : X$, and let $q : a = a$ be the path corresponding to the nonidentity equivalence $e : \mathbf{2} \simeq \mathbf{2}$ defined by $e(0_{\mathbf{2}}) :\equiv 1_{\mathbf{2}}$ and $e(1_{\mathbf{2}}) :\equiv 0_{\mathbf{2}}$. We would like to apply Lemma 4.1.2 to build an f. By definition of X, equalities in subset types (original section 3.5), and univalence, we have $(a = a) \simeq (\mathbf{2} \simeq \mathbf{2})$, which is a set, so (i) holds. Similarly, by definition of X and equalities in subset types we have (ii). Finally, original exercise 2.13 implies that every equivalence $\mathbf{2} \simeq \mathbf{2}$ is equal to either $\mathsf{id}_{\mathbf{2}}$ or e, so we can show (iii) by a four-way case analysis.

Thus, we have $f : \prod_{(x:X)}(x = x)$ such that $f(a) = q$. Since e is not equal to $\mathsf{id}_{\mathbf{2}}$, q is not equal to refl_a, and thus f is not equal to $\lambda x.\, \mathsf{refl}_x$. Therefore, $\prod_{(x:X)}(x = x)$ is not a mere proposition. □